



3D documentation certification methodology

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The 3D documentation certificate records how the scan was made (capture trace), what substantiates its scale (verifications) and exactly what it certifies (file hashes). The app issues it for a finished model; it exists as certificate.json in the .f3d package and as a PDF for the case file. The certificate computes nothing — every value in it comes from the session manifest or from a file hash.

The certificate is not a metrological calibration of the device and does not replace expert judgement. It states by how many independent routes the model scale is substantiated, and with what result.

1. Sources of scale and their independence

Source	What it measures	Independent of
LiDAR depth (time-of-flight)	distance of every image point from the camera; photogrammetry receives a depth prior, the model is built in metres	photogrammetry, the operator
Camera trajectory fit (ARKit → model)	similarity transform between camera positions from ARKit (IMU + visual odometry, metric) and poses from photogrammetry; the fit scale should be ≈ 1.000	LiDAR (works without it), the target
Reference	the operator measures a segment of known length in the model (100 mm target scale, ruler) and enters the true value	LiDAR, ARKit; depends on the operator
T2 target — metrology	the centres of four QR fiducials (A 40 mm, B–D 20 mm) are computed in 3D from LiDAR depth and the camera pose; 6 distances (48–127 mm) are compared with the printed reference	photogrammetry, the operator; verifies the LiDAR + ARKit chain

The T2 target is intended for validation of the app and the device: the same target, the same procedure, multiple scans and multiple devices → the error distribution (mean, maximum, repeatability $\pm$ mm per fiducial) is directly material for a validation study. The reference distances are checked with callipers after printing (centres B–D = 48.00 mm, A–D = 126.82 mm).

2. Documentation class

Class	Conditions	Meaning
A — metrologically verified	LiDAR depth for ≥ 90 % of frames and independent verification (reference or T2 target) with error ≤ 1 % and (trustworthy trajectory fit with scale within ± 1 % or both reference and T2 target)	measurements substantiated by two independent routes; values binding within the declared accuracy
B — metric	LiDAR depth for ≥ 90 % of frames, trustworthy trajectory fit with scale within ± 3 % (or fit not available), independent verification missing or ≤ 2 %	the model is in metres, a second proof is missing — state “no reference” in the report
C — indicative scale	scale known (LiDAR or a coefficient from the trajectory without LiDAR) and verification ≤ 5 %	values indicative; add a reference before citing in an expert opinion
D — unverified scale	scale not substantiated (no LiDAR and no trustworthy fit) or verification > 5 %	illustrative documentation only

“Trustworthy fit” = residual / scale < 150 mm, ≥ 8 matched cameras, scale within (0.05; 20). “Independent verification” = the smaller of the reference and T2 target errors; the T2 target counts when it has ≥ 3 distances and every fiducial ≥ 5 observations (after removing outliers > 25 mm from the median). The class is computed by the same function in the report and in the certificate, so the two documents are

always consistent.

3. Capture trace

From the frame metadata (camera pose, time, depth) the certificate states: start and end, duration, number of frames and frames with depth, resolution, coverage (azimuths of 36, elevation bands of 3), camera path length, range and mean of the camera distance from the scan centre, number of frames with a validly found target. The PDF shows a top view: camera path, frames, scan centre and the 3D fiducial positions. The trace shows that the scan was captured according to the procedure (three heights, full circle, appropriate distance) — and exposes one that was not.

4. Certified content and integrity

- SHA-256 of the model (model/model.usdz), the room mesh (model/room.usda) and the findings (annotations.json) at the time of issue; number of findings and measurements.
- SHA-256 of the archive package, if the scan was archived.
- payloadSHA256 = SHA-256 of the canonical certificate bytes (sorted keys, ISO 8601 dates, no whitespace). Any change to the certificate invalidates the hash.
- Signature (Pro): ECDSA P-256 / SHA-256 with the device key (Secure Enclave where available) over the canonical bytes; the certificate carries the public key (X9.63), the signature (DER), time, signer, institution and the key fingerprint (first 8 bytes of the SHA-256 of the public key).
- Identifier F3D-XXXXX-XXXXX = SHA-256(session ID + model hash + issue time). QR code in the PDF: F3D:<ID>:<class>:<model hash prefix>:<session ID>.

5. Verifying a certificate

- 1 In the app: import the .f3d package. The import verifies the hashes of all files, the package signature (if any) and the certificate: content integrity, the match of the model hash with the model in the package and the certificate signature. The result is in the session overview (“Certificate in package — valid”) and in the report.
- 2 Without the app: `unzip -p scan.f3d model/model.usdz | shasum -a 256` must give the hash from the certificate; `unzip -p scan.f3d certificate.json` contains the certificate, its payloadSHA256 and the signature; any ECDSA P-256 library verifies the public key and signature over the canonical bytes.
- 3 Compare the QR code in the PDF with the ID and hash in the package.

6. Device validation — recommended procedure for a study

- 1 Print the T2 target at 100 %, check the centres B–D (48.00 mm) and the scale (100 mm) with callipers; a print deviation > 0.2 mm → new print.
- 2 Target flat, matte background, diffuse light. Capture ≥ 10 scans with the Object profile, extent Detail, distance 0.3–0.6 m, three heights, on each device.
- 3 From each scan record from the report/certificate: mean and maximum T2 target error (mm, %), scale (k), fiducial repeatability (± mm), trajectory fit scale, reference deviation.
- 4 Evaluate mean, SD and the 95th percentile of errors by device and distance; the class A criterion (≤ 1 %) should be met in ≥ 95 % of scans.
- 5 Validate larger dimensions (body length) with a 500–1000 mm reference (measuring tape, two marks) — reference in the report.

7. Limitations

- LiDAR depth has a resolution of 256×192 px; the fiducial centre accuracy is $\sim 1\text{--}3$ mm at 0.5 m and decreases with distance and oblique views. Hence the median over many frames and outlier removal.
- The T2 target verifies the LiDAR + ARKit chain; the scale of the photogrammetric model is verified by the trajectory fit and the reference. The certificate reports both routes separately.
- The colour patches of the target are indicative; colour-binding documentation requires a commercial ColorChecker.
- The certificate confirms the state at the time of issue; later findings and measurements change the findings hash — issue a new certificate after edits.